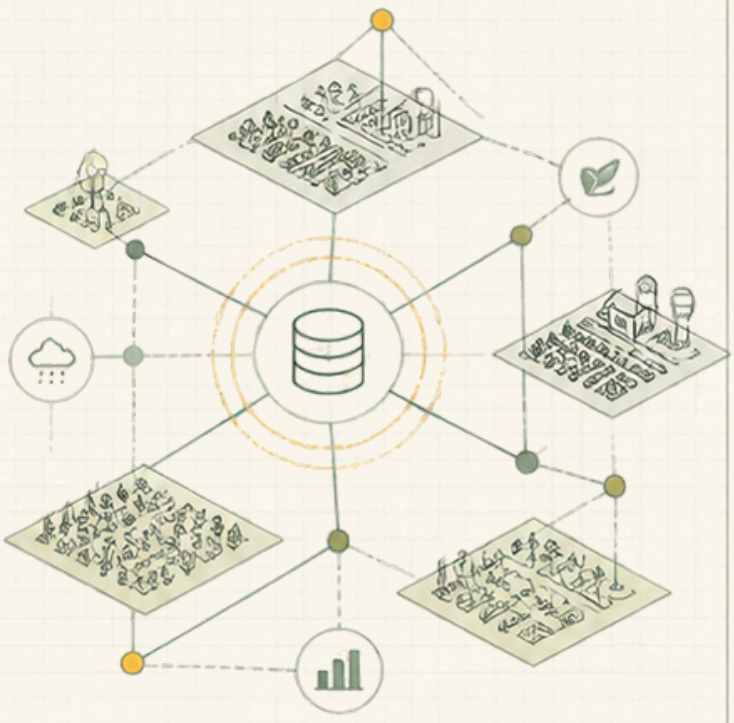
MEKO

official logo asset
to be inserted

MEKO FEATURE PROPOSAL — KISAAN SAATHI

Shared memory and contextual coordination for farm-focused AI agents



Prepared by

Team Spidy

EXECUTIVE SUMMARY

Kisaan Saathi is designed as an agricultural intelligence platform with multiple specialist agents serving farmers across planning, advisory, operations, and oversight. The proposed Meko integration gives those agents controlled access to shared context so they can coordinate, build on prior interactions, and deliver continuity without asking farmers the same questions again.

This is a PROPOSED feature, not an implemented or verified integration.

WHY SHARED CONTEXT MATTERS

- Isolated agents repeat farmer questions, creating friction and fatigue.
- Context is fragmented across interactions and channels.
- Specialist agents can make inconsistent decisions when they cannot see relevant prior context.
- Farmers lose continuity and trust when advice changes without clear reason.
- Operational facts and conversational context can be confused, leading to misinterpretation and errors.

THE PROPOSAL

Meko is proposed as a shared context memory integration accessible through MCP.

MCP is the secure access **protocol/interface** and is not itself a memory store.

YugabyteDB remains the operational structured source of truth.

Design objective: make relevant, permissioned context available to the right specialist agent at the right time while preserving source authority, traceability, human oversight, and safe uncertainty.

FIVE PRINCIPLES

Persistent context

obtain useful approved context across agent interactions.

Scoped access

obtain useful farmer, farm, crop, season, and task-relevant context.

Explicit permissions

consent-aware, tenant-isolated read/write rules.

Traceability

source, timestamp, confidence, version, and retrieval provenance.

Human oversight

agonomy review and escalation remain part of the quality gate.

PROPOSED / VERIFICATION NOTE

Meko APIs, exact product capabilities, performance, and integration status are not assumed and must be verified during implementation.

IMPLEMENTATION, SAFEGUARDS & VALIDATION

PROPOSED IMPLEMENTATION PLAN (6 PHASES) — UNTIL VERIFIED

	PHASE	DELIVERABLE	ACCEPTANCE FOCUS
1.		1. Context schema farmer_id / farm_id, crop and season, source, timestamp, confidence, permissions, version	Schema defined, persistered and ready for use
2.		2. MCP access contracts authenticated client, tool/resource contracts, scope checks, error handling	Secure contracts, ver silenced, errors handled
3.		3. Read/Write memory workflows retrieve relevant content, approve writes, attach provenance, prevent unsupported writes	Reads relevant, writes approved, provenance attached, unsupported writes
4.		4. LangGraph integration explicit context retrieval, reviewer, PASSHOLD/ESCALATE, and persistence nodes	Graph wired, reviewer active, decisions persisted
5.		5. Durable workflow state checkpointer/databases for assumsable failure-mode, state, separated from Meko memory and YugabyteDB facts	State durable, resumable, separation verified
6.		**6. Tests, monitoring & rollout** unit, integration, tenancy, contradiction, failure-mode, latency, audit, staged pilot and rollback checks	Tests pass, monitoring active, pilot validated, rollback ready

All phases are **PROPOSED** until verified through implementation and testing.

SAFEGUARDS — UNTIL VERIFIED

	SAFEGUARD	OUR APPROACH
	Tenancy isolation	no cross-farmer retrieval
	Permission model	farmer/farm/task-scoped access and consent-aware writes
	Sensitive data minimisation	do not store secrets or unnecessary sensitive data
	Redaction	redact sensitive records before persistence or retrieval where required
	TTI / versioning / forget / delete	support these only if available in the verified Meko integration
	Meko unavailable	continue only on available label missing, contact never invent memory
	Source authority	current operational facts come from YugabyteDB, not memory
	Human oversight	hold or escalate when evidence, units, fit, confidence, contradictions, missing data, or safety checks fail

EVALUATION METRICS (NOT CURRENT RESULTS)

	METRIC	WHAT ITVALIDATES
	retrieval relevance	retrieved context is pertinent to the task
	context freshness	reency and currency of retrieved context
	cross-agent consistency	agents reason consistently on shared context
	wrong-scope retrieval rate	retrieval out of data tenant/farm, scope is rare
		writes are accurate, approved, and probably linked
	latency write correcteness	end-to-end time within acceptable thresholds
	token savings	fewer tokens used via better content reuse
	unsupported claim rate	unsupported claims are minimized
	reviewer hold/escalation rate	quality gates effectively catch issues

These are evaluation measures, not current results.

RISKS & LIMITATIONS — WITH MITIGATIONS

	RISK / LIMITATION	MITIGATION
⚠	stale or incorrect memory	timestamps, confidence, versioning, reviewer approval
⚠	**scope leakage**	tenant and farm scope enforcement plus adversarial tests
⚠	**conflicting sources**	YugabyteDB authority and contradiction checks
⚠	**Meko outage**	DB-only fallback with uncertainty label
⚠	**unsupported capabilities or APIs**	verify before implementation and keep adapter boundaries
⚠	**overreliance on memory**	require source citations and current-fact retrieval

Meko supplies shared context.
YugabyteDB remains the operational source of truth.
MCP is the secure access protocol.
LangGraph orchestrates agent collaboration.

Expected benefits are hypotheses to **validate, not measured claims:**
fewer repeated questions, more consistent cross-agent **reasoning.**
clearer provenance, and safer continuity **for farmers.**

PROPOSED — VERIFY APIS, CAPABILITIES, PERFORMANCE, AND INTEGRATION STATUS BEFORE CLAIMING IMPLEMENTATION.